$G \leq GL(n,\mathbb{C})$ matrix Lie group $\xrightarrow{\text{induces}}$ $\mathfrak{g}$ Lie algebra $= \{ X \in M_n(\mathbb{C}) \mid e^{tX} \in G \ \forall t \in \mathbb{R} \}$

$\exp: \mathfrak{g} \to G$ lets us go from $\mathfrak{g}$ to G

- not injective (in general) $\to$ $e^{i2\pi} = e^0$ (for $U(1)$)
- not surjective (in general)

$\to$ if $G = GL(n,\mathbb{C})$ then $\exp: \mathfrak{gl}(n,\mathbb{C}) \to GL(n,\mathbb{C})$ is surjective

$\underline{n=1}$ $\mathfrak{gl}(n,\mathbb{C}) \equiv \mathbb{C}$ $GL(n,\mathbb{C}) = \mathbb{C} \setminus \{0\}$

<u>Prop:</u> if G is a <u>connected</u> MLG, and $g \in G$. Then there are finitely many $X_1, \dots, X_k \in \mathfrak{g}$ s.t. $g = e^{X_1} e^{X_2} \dots e^{X_k}$.

<u>Homomorphism</u>

<u>Def</u>: if G, H are MLG, $\Phi: G \to H$ is a MLG homomorphism if Φ is a group homomorphism and it is smooth

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<u>Def</u>: if $\underline{\mathfrak{g}}, \underline{\mathfrak{h}}$ are Lie algebras, $\phi: \mathfrak{g} \to \mathfrak{h}$ is a Lie algebra homomorphism if

- ϕ is linear (real linear)
- $[\phi(x), \phi(y)] = \phi([x,y])$

<u>Prop:</u> if G, H are MLG with Lie algebras $\underline{\mathfrak{g}}, \underline{\mathfrak{h}}$ and $\underline{\mathfrak{g}} \cong \underline{\mathfrak{h}}$

Then G, H are locally isomorphic.

<u>ex:</u> $G = (\mathbb{R}, +) \cong \left\{ \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \middle| \, x \in \mathbb{R} \right\}$

① $\begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & y \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & x+y \\ 0 & 1 \end{pmatrix}$ $\qquad \phi : x \in \mathbb{R} \mapsto \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}$ is group isomorphism

② $(\mathbb{R}, +)$ think of as group of translations in 1D

if $x \in \mathbb{R}$, $a \in \mathbb{R}$ $\qquad a \triangleright x = x + a$

encode x in a 2D vector $\begin{pmatrix} x \\ 1 \end{pmatrix}$

action of a on $\begin{pmatrix} x \\ 1 \end{pmatrix}$ is $a \triangleright \begin{pmatrix} x \\ 1 \end{pmatrix} = \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ 1 \end{pmatrix} = \begin{pmatrix} x+a \\ 1 \end{pmatrix}$

in general, affine trasformation on $\mathbb{R}^n$ is encoded with an $(n+1)\times(n+1)$ matrix $\begin{pmatrix} A & b \\ 0 & 1 \end{pmatrix}$ $\qquad A \in M_n(\mathbb{R}) \to$ linear part

$b \in \mathbb{R}^n \to$ translation part

action on $x \in \mathbb{R}^n$ is $\begin{pmatrix} A & b \\ 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ 1 \end{pmatrix} = \begin{pmatrix} Ax + b \\ 1 \end{pmatrix}$

<u>back to ex:</u> $\qquad G = (\mathbb{R}, +) \equiv \left\{ \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \,\middle|\, x \in \mathbb{R} \right\}$

$\mathfrak{g} = \left\{ \begin{pmatrix} 0 & x \\ 0 & 0 \end{pmatrix} \,\middle|\, x \in \mathbb{R} \right\}$

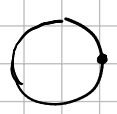
$$e^{\begin{pmatrix} 0 & x \\ 0 & 0 \end{pmatrix}} = \mathbb{1} + \begin{pmatrix} 0 & x \\ 0 & 0 \end{pmatrix} + O(\dots)^2 = \mathbb{1} + \begin{pmatrix} 0 & x \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}$$

$\exp : \mathfrak{g} \to G$ is injective and surjective

$H = U(1) \qquad \underline{h} = \{ i\theta \mid \theta \in \mathbb{R} \}$

$\underline{\mathfrak{g}} \cong \underline{h} \qquad \phi : \begin{pmatrix} 0 & x \\ 0 & 0 \end{pmatrix} \mapsto ix$

$H \ncong G$ $\qquad$ locally : remove $e^{i\pi}$ from $U(1)$, $H \setminus \{e^{i\pi}\} \cong$



$\theta \in [0, 2\pi) \to e^{i\theta}$

<u>Prop:</u> $G \cong H$ then $\underline{\mathfrak{g}} \cong \underline{h}$

Prop: if $\Phi : G \to H$ is a MLG homomorphism, it induces a Lie algebra homomorphism $\phi : \mathfrak{g} \to \mathfrak{h}$

$X \in \mathfrak{g}$ $\quad \phi(X) = \frac{d}{dt} \Phi(e^{tX}) \Big|_{t=0}$

Prop if $\phi : \underline{\mathfrak{g}} \to \underline{\mathfrak{h}}$ is Lie algebra homomorphism, it induces a local homomorphism $\Phi : \underset{\text{open}}{U \subseteq G} \to H$

if $g_1, g_2, g_1 g_2 \in U$ then $\Phi(g_1 g_2) = \Phi(g_1)\Phi(g_2)$

and $\Phi(e^X) = e^{\phi(X)}$ as long as $e^X \in U$

Prop if $\phi : \underline{\mathfrak{g}} \to \underline{\mathfrak{h}}$ is Lie algebra homomorphism, and G is simply connected then there is a unique $\Phi : G \to H$ MLG homomorphism with

$$\Phi(e^X) = e^{\phi(X)}$$

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ex of breakage

$\Phi : U(1) \to U(1)$ MLG isomorphism

$e^{i\theta} \mapsto e^{i2\theta}$

$\Phi(e^{i\theta} e^{i\sigma})$

$= \Phi(e^{i(\theta+\sigma)}) = e^{i2(\theta+\sigma)}$

$= e^{i2\theta} e^{i2\sigma}$

induces $\phi : \underline{u}(1) \to \underline{u}(1)$

if $i\theta \in \underline{u}(1)$ $\quad \phi(i\theta) = \frac{d}{dt}\left(\Phi(e^{it\theta})\right)\Big|_{t=0}$

$$= \frac{d}{dt}\left(e^{i2\theta t}\right)\Big|_{t=0} = i2\theta$$

$\phi(i\theta) = i2\theta$ $\quad$ ϕ is a Lie algebra isomorphism

start with $\psi : i\theta \in \underline{u}(1) \mapsto \frac{i\theta}{2} \in \underline{u}(1)$ Lie algebra isomorphism

try to define $\Psi : U(1) \to U(1)$ s.t. $\Psi(e^{i\theta}) = e^{i\psi(\theta)} = e^{i\theta/2}$

this cannot be extended to the whole group because

$e^{i2\pi} = 1$ but $\psi(e^{i2\pi})$ would be $e^{i\pi} = -1$

<u>other example</u> $G = \mathbb{R} = \left\{ \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \mid x \in \mathbb{R} \right\}$ $\mathfrak{g} = \left\{ \begin{pmatrix} 0 & x \\ 0 & 0 \end{pmatrix} \mid x \in \mathbb{R} \right\}$

$\phi : \begin{pmatrix} 0 & x \\ 0 & 0 \end{pmatrix} \in \mathfrak{g} \mapsto \begin{pmatrix} 0 & \alpha x \\ 0 & 0 \end{pmatrix} \in \mathfrak{g}$ $\alpha \in \mathbb{R}$

$\Phi\left(e^{\begin{pmatrix} 0 & x \\ 0 & 0 \end{pmatrix}}\right) = e^{\phi\begin{pmatrix} 0 & x \\ 0 & 0 \end{pmatrix}} = e^{\begin{pmatrix} 0 & \alpha x \\ 0 & 0 \end{pmatrix}} = \begin{pmatrix} 1 & \alpha x \\ 0 & 1 \end{pmatrix}$ is MLG isomorphism

$\forall \alpha$

<u>Reps</u>

- if G is a MLG, a representation of G on a finite-dimensional complex vector space V is a smooth map $\rho : G \to GL(V)$

  s.t. $\rho(g_1 g_2) = \rho(g_1)\rho(g_2)$
  $\rho(e) = \mathbb{1}$

  $GL(V) = \{$ invertible linear maps $V \to V \}$

- if $\underline{g}$ is a Lie algebra, a representation of $\underline{g}$ on a finite-dimensional complex vector space V is a linear map $\pi : \underline{g} \to \underline{gl}(V)$

  s.t. $\pi([x,y]) = [\pi(x), \pi(y)]$

  $\underline{gl}(V) = \{$ linear maps $V \to V \}$

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if $\rho : G \to GL(V)$ rep. then $\pi : \mathfrak{g} \to \mathfrak{gl}(V)$ with $\pi(x) = \frac{d}{dt}\left(\rho(e^{tx})\big|_{t=0}\right)$ is a Lie algebra rep.